Physics IGCSE	Summary formula	Physics grade 11
Content	**Symbols**	**System international unit (SI unit)**
1. Electric charge on an object which is excesses e^- **2.** Electric charge on an object which is lost e^- **3.** The Coulomb's Law	1. $q=-n\|e\|$ 2. $q=+n\|e\|$ 3. $F_{1\to2}=F_{2\to1}=k\frac{\|q_1\|\|q_2\|}{r^2}$	q are the electric charges (C). n is the number of electrons. $e=1.6\times10^{-19}C$ is the charge on one electron. $k=9\times10^9\ Nm^2/C^2$ is the constant of proportionality. r is the distance between both objects measured in (m). $F_{1\to2}=F_{2\to1}$ is the electric force measured in (N).
4. Electric field exists around ang charged object. **5.** Electric force on a tested charges. **6.** Electric potential difference. **7.** Work done and Potential difference. **8.** Electric potential difference in uniform field. **9.** The capacitance of capacitor.	4. $E=\frac{F_{on\cdot q'}}{q'},or\quad E=k\frac{\|q\|}{r^2}$ 5. $F_e=F_{on\cdot q'}=q'\|E\|$ 6. $V=\frac{PE}{q}$ 7. $W_{on\cdot q}=-q\Delta V,\Delta V=V_1-V_2$ 8. $\Delta V=Ed,E=\frac{\Delta V}{d}$ 9. $C=\frac{q}{V},q=CV$	r is the distance between points of filed and electric charges (m). F is the electric force on test charges (N). E is the electric field (N/C). PE is the potential energy (J) V is the electric potential difference (V). ΔV is the potential difference (V). E is the uniform field (V/m) d is the distance moved by a charge (m). C is the capacitance (F).
10. Angular velocity **11.** Angular acceleration **12.** Torque **13.** Net torque **14.** Center of mass **15.** Conditions for equilibrium **16.** Centrifugal acceleration **17.** Linear momentum **18.** Impulse **19.** Momentum and impulse theorem **20.** Angular momentum **21.** Angular impulse-angular momentum **22.** The law of conservation angular momentum	10. $\omega=\frac{\Delta\theta}{\Delta t}=\frac{\theta_2-\theta_1}{t_2-t_1}$ 11. $\alpha=\frac{\Delta\omega}{\Delta t}=\frac{\omega_2-\omega_1}{t_2-t_1}$ 12. $\tau=Fr\sin\theta$ 13. $\tau_{net}=I\alpha$ 14. $x_{cm}=\sum_{i=1}^{n}\frac{m_ix_i}{m_i}$ 15. $\begin{cases}\sum F=0\\ \sum\tau=0\end{cases}$ 16. $a_c=\frac{v^2}{r},v=r\omega$ 17. $p=mv$ 18. $I=F\Delta t$ 19. $F\Delta t=m\Delta v\Rightarrow F=\frac{m\Delta v}{\Delta t}$ 20. $L=I\omega$ 21. $\tau\Delta t=L_f-L_i$ 22. $L_f=L_i\Leftrightarrow I_i\omega_i=I_f\omega_f$	ω is the angular velocity (rad/s). θ is the angular displacement (rad). Δt is the time taken (s). α is the angular acceleration (rad/s^2) τ is torque $(N.m)$. F is the force (N). r is the effort's arm length (m) I is the moment of inertia $(kg.m^2)$. x_{cm} is the center of mass through x-axis (m,cm) m is the mass of a system (kg). a_c is the centrifugal acceleration (m/s^2). v is the uniform velocity (m/s). r is the radius of a circular path (m). p is the momentum $(kg.m/s)$. $\Delta v=v_2-v_1$ is the change in velocity (m/s). Δt is the time taken (s). L is the angular momentum $(kg.m^2/s)$.

23. Law of conservation of momentum **24.** Conservation of energy in perfectly elastic collision	23. $m_1v_{1i}+m_2v_{2i}=m_1v_{1f}+m_2v_{2f}$ 24. $\frac{1}{2}m_1v_{1i}^2+\frac{1}{2}m_2v_{2i}^2=\frac{1}{2}m_1v_{1f}^2+\frac{1}{2}m_2v_{2f}^2$	v_{1i}, v_{2i} are velocities before impact (m/s). v_{1f}, v_{2f} are velocities after impact (m/s). m_1, m_2 are masses (kg).

REVIEW FOR FINAL EXAM GRADE 11

A. Multiple choice questions

1. What is the effect of a negative charge on a positive charge?
a. Attract **b.** Repel
c. Moving at right angles upward **d.** None of the above

2. What are the charge carriers in a metal?
a. Proton **b.** Neutrons
c. Electrons **d.** Ions

3. Which of the following is NOT a unit for angle?
a. Degree **b.** Radian
c. Revolution **d.** Revolution per minute

4. Which of the following is the correct units for electric charge?
a. $N.m^2/C^2$ **b.** N/C
c. V/m **d.** C

5. The magnitude of the electric force between two points of charge is F. If the distance between the charges is double, the electrostatic force between the charge will become
a. F/4 **b.** F/2
c. 4F **d.** 2F

6. A change or transfer of momentum is known as
a. Restoring force **b.** Conversion of energy
c. Force **d.** Impulse

7. What is the angular velocity of $\omega = 635 rpm$ in rad/s?
a. $\omega = 66.49\, rad/s$ **b.** $\omega = 3989.82\, rad/s$
c. $\omega = 2190\, rad/s$ **d.** $\omega = 3810\, rad/s$

8. When you double the charge on both particles in a pair, what effect does this have on the force between them?
a. 2 times **b.** 3 times
c. 4 times **d.** 5 times

9. What is the unit of torque?
a. J/s **b.** $v = kg.m/s$
c. $kg.m^2/s$ **d.** $N.m$

10. Moment of inertia of a solid sphere of mass M and radius R is
a. $\frac{1}{2}MR^2$ **b.** $\frac{2}{3}MR^2$ **c.** $\frac{2}{5}MR^2$ **d.** MR^2

11. A flywheel of diameter 1.2m has a constant angular acceleration of 5 rad/s^2. The tangential acceleration of a point on its rim is
a. 5.0 rad/s^2 **b.** 3.0 m/s^2
c. 6.0 m/s^2 **d.** 5.0 m/s^2

12. If a wheel turning at a constant rate completes 100 revolutions in 10 seconds, its angular speed is
a. 62.8 rad/s **b.** 0.31 rad/s
c. 0.95 rad/s **d.** 0.105 rad/s

13. Electric field strength can be defined as
a. E=Q/F **b.** E=W/F
c. E=P/Q **d.** E=F/Q

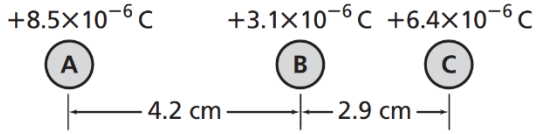
14. Which of the following is a conductor?
a. Glass **b.** Tungsten
c. Wood **d.** Plastic

15. The electric force between two charged object depends on which of the following?
a. their masses and their separation.
b. their speeds.
c. their charge and their separation.
d. their masses and their charge.

16. How many electrons have been removed from a positively charged electroscope if it has a net charge of $7.5\times10^{-11} C$?
a. $7.5\times10^{-11} elctrons$ **b.** $2.1\times10^{-9} elctrons$
c. $1.2\times10^{8} elctrons$ **d.** $4.7\times10^{8} elctrons$

17. The force exerted on a particle with a charge of $5.0\times10^{-9} C$ by a second particle that is $4cm$ away is $8.4\times10^{-5} N$. What is the charge of the second particle?
a. $4.2\times10^{-13} C$ **b.** $3.0\times10^{-9} C$
c. $2.0\times10^{-9} C$ **d.** $6.0\times10^{-5} C$

18. Three charges, A, B, and C, are located in a line, as shown below. What is the net on charge B?
a. 78 N toward A **b.** 130 N toward A
c. 78 N toward C **d.** 210 N toward C



19. What is the charge on an electroscope that has an excess of 4.8×10^{10} electrons?
a. $3.3\times10^{-30} C$ **b.** $7.7\times10^{-9} C$
c. $4.8\times10^{-10} C$ **d.** $4.8\times10^{10} C$

20. Two charged bodies exert a force of 86 N on each other. If they are moved so that they are six times further apart, what is the new force that they will exert on each other?
a. 2.4 N **b.** 86 N
c. 14 N **d.** 5.2 × 10^2 N

21. Two equally charged bodies exert a force of 90 N on each other. If one of the bodies is exchanged for a body of the same size, but three times as much charge, what is the force that they will exert on each other?

a. 10 N **b.** 2.7×10^2 N
c. 30 N **d.** 8.1×10^2 N

22. An alpha particle has a mass of 6.68×10^{-27} kg and a charge of 3.2×10^{-19} C. What is the ratio of the electrostatic force to the gravitational force between tow alpha particles?

a. 1 **b.** 2.3×10^{15}
c. 4.8×10^7 **d.** 3.1×10^{35}

23. Charging a neutral body by touching it with a charged body is called charging by

a. Conduction **b.** Grounding
c. Induction **d.** Discharging

24. Macy rubs a balloon with wool, giving the balloon a charge of -8.9×10^{-14} C. What is the force between the balloon and a metal sphere that is charged to 25 C and is 2 km away?

a. 8.9×10^{-15} N **b.** 2.2×10^{-12} N
c. 5.0×10^{-9} N **d.** 5.6×10^4 N

25. A neutral rubber rod is rubbed with fur and acquires a charge of $-2.0\times10^{-6}C$. The charge on the fur is: $Q1 = -Q2$

a. $-2.0\times10^{-6}C$ **b.** $+2.0\times10^{-6}C$
c. $-1.0\times10^{-6}C$ **d.** $+4.0\times10^{-6}C$

26. A charge of 0.2×10^{-9} C exerts a repulsive force of 0.2 N on an unknown charge located 0.30m away.

a. 0.01 C **b.** 0.02 C
c. 0.005 C **d.** 0.1 C

27. A wheel initially turning at 2000 rpm uniformly increases its speed to 6600 rpm in 9.00 sec. Calculate the angular acceleration of the wheel in rad/s^2 and the number revolutions turned by the wheel during the 9.00 sec interval.

a. 53.49 rad/s^2, 645.65 rev **b.** 51.11 rad/s^2, 20700 rev
c. 511.11 rad/s^2, 3700 rev **d.** 53.49 rad/s^2, 119 rev

B. Critical thinking questions

28. What property makes metal a good conductor and rubber a good insulator?

29. If you wipe a compact disc with a clean cloth, why does the CD then attract dust?

30. Explain how to charge a conductor negatively if you have only a positively charged rod.

31. What happens to the gravitational force between two objects if the distance between them is halved?

32. An object moves in a circular path with constant speed. What type of acceleration does it experience? Explain its direction and how it is calculated.

33. A satellite orbits Earth at a fixed altitude. Explain how the centripetal force acting on the satellite is related to its orbital velocity and the gravitational pull of Earth.

C. Work problems

34. A wheel of 50cm in diameter turning at 150 rpm increases its frequency to 600 rpm in 5.0 s. Find
(a) the constant angular acceleration
(b) the tangential acceleration of a point on its rim.

35. A toy consisting of two small spheres, each of mass 0.6 kg is attached to the ends of a 0.5 m-long, lightweight rod. Find the moment of inertia of the toy about the center of the rod.

36. In an experiment using Coulomb's apparatus, a sphere with a charge of 3.6×10^{8} C is 1.4 cm from a second sphere of unknown charge. The force between the spheres is 2.7×10^{2} N. What is the charge of the second sphere?

37. A small sphere of charge 2.4 C experiences a force of 0.36 N when a second sphere of unknown charge is placed 5.5 cm from it. What is the charge of the second sphere?

38. A positive test charge of 5.0×10^6 C is in an electric field that exerts a force of 2.0×10^4 N on it. What is the magnitude of the electric field at the location of the test charge?

39. A negative charge of 2.0×10^8 C experiences a force of 0.060 N to the right in an electric field. What are the field's magnitude and direction at that location?

40. A pith ball weighing 2.1×10^3 N is placed in a downward electric field of 6.5×10^4 N/C. What charge (magnitude and sign) must be placed on the pith ball so that the electric force acting on it will suspend it against the force of gravity?

41. A positive charge of 3.0×10^7 C is located in a field of 27 N/C directed toward the south. What is the force acting on the charge?

42. What is the magnitude of the electric field strength at a position that is 1.2 m from a point charge of 4.2×10^6 C?

43. The electric field that is 0.25 m from a small sphere is 450 N/C toward the sphere. What is the charge on the sphere?

44. How far from a point charge of 2.4×10^6 C must a test charge be placed to measure a field of 360 N/C?

45. The electric field intensity between two large, charged parallel metal plates is 6000 N/C. The plates are 0.05 m apart. What is the electric potential difference between them?

46. The electric field in a particle-accelerator machine is 4.5×10^5 N/C. How much work is done to move a proton 25 cm through that field?

47. An oil drop weighs 1.9×10^{15} N. It is suspended in an electric field of 6.0×10^3 N/C. What is the charge on the drop? How many excess electrons does it carry?

48. A 27 μF capacitor has an electric potential difference of 45 V across it. What is the charge on the capacitor?

49. Two freight cars, each with a mass of 3.0×10^5 kg, collide and stick together. One was initially moving at 2.2 m/s, and the other was at rest. What is their final speed?

50. A 0.105-kg hockey puck moving at 24 m/s is caught and held by a 75-kg goalie at rest. With what speed does the goalie slide on the ice?

51. A 35.0-g bullet strikes a 5.0-kg stationary piece of lumber and embeds itself in the wood. The piece of lumber and bullet fly off together at 8.6 m/s. What was the original speed of the bullet?

52. The bullet in the previous problem strikes a 2.5-kg steel ball that is at rest. The bullet bounces backward after its collision at a speed of 5.0 m/s. How fast is the ball moving when the bullet bounces backward?

53. A 0.50-kg ball that is traveling at 6.0 m/s collides head-on with a 1.00-kg ball moving in the opposite direction at a speed of 12.0 m/s. The 0.50-kg ball bounces backward at 14 m/s after the collision. Find the speed of the second ball after the collision.

54. A 4.00-kg model rocket is launched, expelling 50.0 g burned fuel from its exhaust at a speed of 625 m/s. What is the velocity of the rocket after the fuel has burned? Hint: Ignore the external forces of gravity and air resistance.

55. A thread holds a 1.5-kg cart and a 4.5-kg cart together. After the thread is burned, a compressed spring pushes the carts apart, giving the 1.5-kg cart a speed of 27 cm/s to the left. What is the velocity of the 4.5-kg cart?

56. A stationary billiard ball, with a mass of 0.17 kg, is struck by an identical ball moving at 4.0 m/s. After the collision, the second ball moves 60.0° to the left of its original direction. The stationary ball moves 30.0° to the right of the moving ball's original direction. What is the velocity of each ball after the collision?

57. A 0.145-kg baseball is pitched at 42 m/s. The batter hits it horizontally to the pitcher at 58 m/s.

a. Find the change in momentum of the ball.

b. If the ball and bat are in contact for 4.6×10^{-4} s, what is the average force during contact?

58. Bowling A force of 186 N acts on a 7.3-kg bowling ball for 0.40 s. What is the bowling ball's change in momentum? What is its change in velocity?

59. A 0.150-kg ball, moving in the positive direction at 12 m/s, is acted on by the impulse shown in the graph in Figure 9-16. What is the ball's speed at 4.0 s?

60. If a truck has a linear acceleration of 1.85 m/s^2 and the wheels have an angular acceleration of 5.23 rad/s^2, what is the diameter of the truck's wheels?

61. If a torque of 55.0 Nm is required and the largest force that can be exerted by you is 135 N, what is the length of the lever arm that must be used?

62. Torque A solid steel wheel has a mass of 15 kg and diameter of 0.44 m. It starts at rest. You want to make it rotates at 8.0 rev/s in 15 s.

a. What torque must be applied to the wheel?

b. If you apply the torque by wrapping a strap around the outside of the wheel, how much force should you exert on the strap

63. A 7.3-kg ladder, 1.92 m long, rests on two sawhorses, as shown in Figure 8-15. Sawhorse A, on the left, is located 0.30 m from the end, and sawhorse B, on the right, is located 0.45 m from the other end. Choose the axis of rotation to be the center of mass of t ladder.

a. What are the torques acting on the ladder?

b. Write the equation for rotational equilibrium.

c. Solve the equation for F_A in terms of F_g.

d. How would the forces exerted by the two sawhorses change if A were moved very close to, but not directly under, the center of mass?

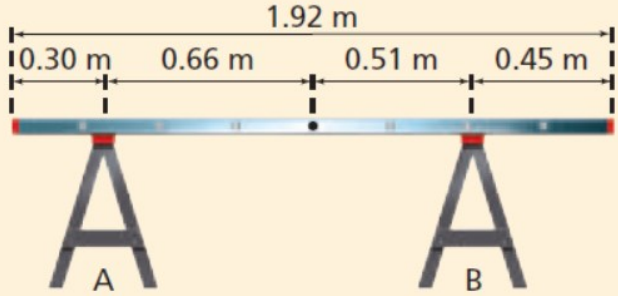


Figure 8-15

64. Propeller A propeller spins at 1880 rev/min.

a. What is its angular velocity in rad/s?

b. What is the angular displacement of the propeller in 2.50 s?

65. A toy consisting of two balls, each 0.45 kg, at the ends of a 0.46-m-long, thin, lightweight rod is shown in Figure 8-25. Find the moment of inertia of the toy. The moment of inertia is to be found about the center of the rod.

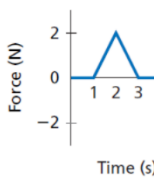


Figure 9-16

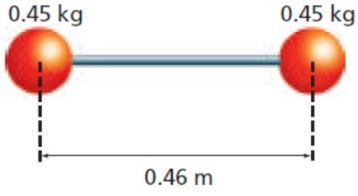


Figure 8-25